

EXACT 2026 Data Pipeline & Disclosure Statement

Team: AI WITH BRO | **Pipeline:** Local ingestion datasets, indexing formats, and self-hosted model details

This document outlines the data sources and ingestion pipelines configured in this repository. All datasets are processed locally using offline python scripts. The environment uses zero commercial API calls or closed-source platforms. The OpenAI-compatible endpoints refer to the API protocol structure running locally via `Llama-server`.

Dataset Registry & Preprocessing

Source	Pipeline Stage	Schema Details	Processing and Engineering Controls	Compliance
EXACT 2026 Dataset	System validation	JSON/CSV task schema	Deterministically parsed by FastAPI handlers; filters out evaluation QA identifiers.	Disclosed
FOLIO Dataset	Code-gen fine-tuning	Natural language premises	Converted to formal Z3 variables and SMT assertions using structured parser scripts.	Disclosed
PhysicsFormulae	Retrieval indexing	Equation constants	Normalized into JSONL document entries mapping physics variables and constants.	Disclosed
Electro Textbook	Code-gen training	Text problems	Distilled into structured code templates and SymPy examples; excluded unless source list is attached.	Disclosed
RAG Database	Runtime retrieval	JSONL formula vectors	Built via local sentence-transformers; falls back to raw code-generation if missing.	Disclosed

Data Pipeline Controls

The system implements strict guidelines to enforce data isolation and safety:

- **Sanitized Parsing:** Input values are cleaned of escape sequences and script wrappers prior to pipeline execution.
- **Offline Indexing:** Retrieval databases (BM25 indexes and vector databases) are compiled locally on the server during the build phase.
- **No Cloud Dependencies:** Training, dataset indexing, retrieval, and inference are configured to execute in air-gapped environments.
- **Verification Suite:** Evaluates model weights against schema validation files to block any malformed output.